Syncner Learning Management System - Complete API Endpoints

Document Information

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Authentication Endpoints

POST /api/auth/register

- **Description**: Register a new user
- **Authentication**: None
- **Request Body**:

```
{
  "name": "string",
  "email": "string",
  "password": "string",
  "careerTrack": "string"
}
```

POST /api/auth/login

- **Description**: User login

- **Authentication**: None
- **Request Body**:

```
{
  "email": "string",
  "password": "string"
}
```

POST /api/auth/logout

- **Description**: User logout
- **Authentication**: Required

GET /api/auth/profile

- **Description**: Get current user profile
- **Authentication**: Required

PUT /api/auth/profile

- **Description**: Update user profile
- **Authentication**: Required
- **Request Body**:

```
{
  "name": "string",
  "phone": "string",
  "location": "string",
  "college": "string",
  "department": "string"
}
```

POST /api/auth/forgot-password

- **Description**: Request password reset
- **Authentication**: None
- **Request Body**:

```
{
  "email": "string"
}
```

POST /api/auth/reset-password

- **Description**: Reset password with token

- **Authentication**: None
- **Request Body**:

```
{
  "token": "string",
  "password": "string"
}
```

User Management Endpoints

GET /api/users/profile

- **Description**: Get user profile
- **Authentication**: Required

PUT /api/users/profile

- **Description**: Update user profile
- **Authentication**: Required

GET /api/users/

- **Description**: Get all users (admin only)
- **Authentication**: Required (Admin)

Student Endpoints

GET /api/student/portfolio/me

- **Description**: Get current student's portfolio
- **Authentication**: Required

GET /api/student/portfolio/:studentId

- **Description**: Get specific student's portfolio
- **Authentication**: Required

GET /api/student/portfolio/public/:studentId

- **Description**: Get public portfolio (no auth required)
- **Authentication**: None

Staff Endpoints

GET /api/staff/students

- **Description**: Get all students with filters and pagination
- **Authentication**: Required (Staff/Admin)
- **Query Parameters**:

    - `page`: Page number
    - `limit`: Items per page
    - `search`: Search term
    - `status`: Filter by status
    - `careerTrack`: Filter by career track
    - `sortBy`: Sort field
    - `sortOrder`: Sort direction

GET /api/staff/students/:id

- **Description**: Get detailed student information
- **Authentication**: Required (Staff/Admin)

GET /api/staff/analytics

- **Description**: Get staff analytics data
- **Authentication**: Required (Staff/Admin)

GET /api/staff/leaderboard

- **Description**: Get overall leaderboard
- **Authentication**: Required (Staff/Admin)

GET /api/staff/leaderboard/:careerTrack

- **Description**: Get career track specific leaderboard
- **Authentication**: Required (Staff/Admin)

GET /api/staff/test/:id

- **Description**: Get test details
- **Authentication**: Required (Staff/Admin)

Admin Endpoints

GET /api/admin/users

- **Description**: Get all users with management capabilities
- **Authentication**: Required (Admin)
- **Query Parameters**:
    - `page`: Page number
    - `limit`: Items per page
    - `search`: Search term
    - `role`: Filter by role
    - `status`: Filter by status

POST /api/admin/users

- **Description**: Create new user
- **Authentication**: Required (Admin)

- **Request Body**:

```
{
  "name": "string",
  "email": "string",
  "password": "string",
  "role": "string",
  "careerTrack": "string"
}
```

PUT /api/admin/users/:id

- **Description**: Update user
- **Authentication**: Required (Admin)

DELETE /api/admin/users/:id

- **Description**: Delete user
- **Authentication**: Required (Admin)

POST /api/admin/users/:id/reset-password

- **Description**: Reset user password
- **Authentication**: Required (Admin)
- **Request Body**:

```
{
  "password": "string"
}
```

GET /api/admin/analytics

- **Description**: Get admin analytics
- **Authentication**: Required (Admin)
- **Query Parameters**:
  - `dateRange`: Date range filter

GET /api/admin/level-progress

- **Description**: Get level progress analytics
- **Authentication**: Required (Admin)

GET /api/admin/health

- **Description**: Admin service health check
- **Authentication**: None

GET /api/admin/diagnostics

- **Description**: System diagnostics
- **Authentication**: Required (Admin)

GET /api/admin/check-progress-orders

- **Description**: Check progress order consistency
- **Authentication**: Required (Admin)

POST /api/admin/fix-progress-orders

- **Description**: Fix progress order issues
- **Authentication**: Required (Admin)

POST /api/admin/cleanup-orphaned-progress

- **Description**: Cleanup orphaned progress records
- **Authentication**: Required (Admin)

Curriculum Endpoints

GET /api/curriculum

- **Description**: Get all curriculum items
- **Authentication**: Required (Admin)

POST /api/curriculum

- **Description**: Create curriculum item
- **Authentication**: Required (Admin)
- **Request Body**:

```
{
  "title": "string",
  "description": "string",
  "type": "string",
  "careerTrack": "string",
  "level": "number",
  "order": "number",
  "content": "string",
  "externalLink": "string",
  "options": ["string"],
  "answer": "string",
  "resources": ["string"]
}
```

GET /api/curriculum/:id

- **Description**: Get curriculum item by ID

- **Authentication**: Required (Admin)

PUT /api/curriculum/:id

- **Description**: Update curriculum item
- **Authentication**: Required (Admin)

DELETE /api/curriculum/:id

- **Description**: Delete curriculum item
- **Authentication**: Required (Admin)

PATCH /api/curriculum/:id/toggle

- **Description**: Toggle curriculum item status
- **Authentication**: Required (Admin)

GET /api/curriculum/stats

- **Description**: Get curriculum statistics
- **Authentication**: Required (Admin)

POST /api/curriculum/invalidate-cache

- **Description**: Invalidate curriculum cache
- **Authentication**: Required (Admin)

PUT /api/curriculum/bulk-order

- **Description**: Update bulk order of curriculum items
- **Authentication**: Required (Admin)

DELETE /api/curriculum/level

- **Description**: Delete level by track and order
- **Authentication**: Required (Admin)
- **Query Parameters**:
  - `careerTrack`: Career track name
  - `order`: Level order

PUT /api/curriculum/level/reorder

- **Description**: Reorder levels within track
- **Authentication**: Required (Admin)

GET /api/curriculum/type/:type/track/:careerTrack

- **Description**: Get curriculum by type and track
- **Authentication**: Required

GET /api/curriculum/content/:type

- **Description**: Get content by type
- **Authentication**: Required (Admin)

GET /api/curriculum/content/:type/track/:careerTrack

- **Description**: Get content by type and track
- **Authentication**: Required (Admin)

POST /api/curriculum/content

- **Description**: Create content item
- **Authentication**: Required (Admin)

PUT /api/curriculum/content/:id

- **Description**: Update content item
- **Authentication**: Required (Admin)

DELETE /api/curriculum/content/:id

- **Description**: Delete content item
- **Authentication**: Required (Admin)

Progress Endpoints

GET /api/progress/:careerTrack

- **Description**: Get user progress for career track
- **Authentication**: Required

GET /api/progress/levels-status

- **Description**: Get levels status
- **Authentication**: Required

GET /api/progress/unlocked/:levelOrder

- **Description**: Check if level is unlocked
- **Authentication**: Required

GET /api/progress/test-unlocked/:levelId

- **Description**: Check if test is unlocked
- **Authentication**: Required

GET /api/progress/exercise-unlocked/:levelId

- **Description**: Check if exercise is unlocked
- **Authentication**: Required

GET /api/progress/exercise-links/:levelId

- **Description**: Get exercise links
- **Authentication**: Required

GET /api/progress/level-sections/:levelId

- **Description**: Get level sections status
- **Authentication**: Required

POST /api/progress/complete

- **Description**: Mark level as completed
- **Authentication**: Required
- **Request Body**:

```
{
  "levelId": "string",
  "levelOrder": "number",
  "section": "string"
}
```

POST /api/progress/learn-progress

- **Description**: Update learning progress
- **Authentication**: Required
- **Request Body**:

```
{
  "levelId": "string",
  "timeSpent": "number",
  "linksAccessed": ["string"]
}
```

POST /api/progress/test-attempt

- **Description**: Submit test attempt
- **Authentication**: Required
- **Request Body**:

```
{
  "levelId": "string",
  "score": "number",
  "answers": ["string"]
}
```

POST /api/progress/submit-exercise

- **Description**: Submit exercise
- **Authentication**: Required
- **Request Body**:

```
{
  "levelId": "string",
  "links": ["string"]
}
```

PUT /api/progress/update-exercise

- **Description**: Update exercise submission
- **Authentication**: Required

Training Endpoints

GET /api/training/levels

- **Description**: Get all training levels
- **Authentication**: Required
- **Query Parameters**:
  - `track`: Career track filter

GET /api/training/levels/:id

- **Description**: Get specific training level
- **Authentication**: Required

Attendance Endpoints

POST /api/attendance/mark

- **Description**: Mark attendance
- **Authentication**: Required

GET /api/attendance/:userId

- **Description**: Get user attendance
- **Authentication**: Required

GET /api/attendance/stats/:userId

- **Description**: Get attendance statistics
- **Authentication**: Required

AI Endpoints

POST /api/ai/chat

- **Description**: Chat with AI
- **Authentication**: Required
- **Request Body**:

```
{
  "message": "string",
  "context": "string"
}
```

Settings Endpoints

GET /api/admin/settings

- **Description**: Get platform settings
- **Authentication**: Required (Admin)

PUT /api/admin/settings

- **Description**: Update platform settings
- **Authentication**: Required (Admin)
- **Request Body**:

```
{
  "geminiApiKey": "string",
  "emailSettings": "object",
  "systemSettings": "object"
}
```

Track Endpoints

GET /api/tracks

- **Description**: Get all career tracks
- **Authentication**: Required

GET /api/tracks/:id

- **Description**: Get specific career track
- **Authentication**: Required

Health Check Endpoints

GET /api/health

- **Description**: Main health check endpoint
- **Authentication**: None
- **Response**:

```
{
  "success": true,
  "message": "Server is running",
  "timestamp": "string",
  "uptime": "string",
  "database": "string",
  "gemini": "string",
  "memoryUsageMB": "number"
}
```

GET /api/auth/health

- **Description**: Auth service health check
- **Authentication**: None

GET /api/users/health

- **Description**: Users service health check
- **Authentication**: None

GET /api/tracks/health

- **Description**: Tracks service health check
- **Authentication**: None

GET /api/training/health

- **Description**: Training service health check
- **Authentication**: None

GET /api/admin/health

- **Description**: Admin service health check
- **Authentication**: None

GET /api/admin/settings/health

- **Description**: Settings service health check
- **Authentication**: None

GET /api/ai/health

- **Description**: AI service health check
- **Authentication**: None

GET /api/curriculum/health

- **Description**: Curriculum service health check
- **Authentication**: None

Public Endpoints

GET /api/student/portfolio/public/:studentId

- **Description**: Get public portfolio (no authentication required)
- **Authentication**: None

File Upload Endpoints

POST /api/upload

- **Description**: Upload file
- **Authentication**: Required
- **Content-Type**: multipart/form-data

Error Responses

All endpoints return consistent error responses:

```
{
  "success": false,
  "message": "Error description",
  "error": "Detailed error information"
}
```

Authentication

Most endpoints require JWT authentication. Include the token in the Authorization header:

```
Authorization: Bearer <jwt_token>
```

Rate Limiting

- **Production**: 100 requests per 15 minutes per IP
- **Development**: No rate limiting

CORS

Allowed origins:

- `https://syncner-test.netlify.app`
- `http://localhost:5173`
- `http://localhost:3000`

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